

IoRT enabled gesture mimicking Robotic Hand based on Computer Vision

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Abstract—Human Robot Interaction (HRI) is a multi-model, progressive approach where humans get together with robots to complement one another. The Internet of Robotic Things (IoRT) happens to be an intriguing technology of the modern world, but it becomes immersive when combined with computer vision. IoRT refers to the control of robots over the internet. According to the International Labor Organization (ILO), close to 3 million people die each year due to occupational hazards. Explosives, munition, chemical processing, and mining industries are in the midst of pernicious industries. This paper presents a human gesture-controlled robotic hand that uses MediaPipe and OpenCV libraries to classify gestures and proposes an IoRT architecture. The Cortex M0+ based Raspberry Pi Pico MCU has been used to realize the concept of AI on embedded systems where a worker makes a robotic entity to mimic him while sitting in a safe cabin, eliminating the need for numerous sensors and cumbersome wearables.

Keywords—Human Computer Interaction (HCI), Human Robot Interaction (HRI), Internet of Robotic Things (IoRT), Internet of Things (IoT), Computer Vision, OpenCV, MediaPipe, Gesture, Robotic Hand, Raspberry Pi Pico.

I. Introduction

In the following years, researchers are head-to-head to discover more optimum human-centered channel to interact with computers. An exquisite and potential way of human computer interaction (HCI) is the notion to make computers capable of interpreting human cues, technically referred to as perceptive user interfaces (viz-a-via, behavior, biometric, voice, and vision interface) [1]. Amongst all, vision is probable of fetching profuse information, in a subtle way, and at a minimum cost. Consequently, computer-vision has become a center of attraction for a variety of applications, whether in healthcare, industry, agriculture [2], transportation or autonomous cars due to its versatility. Common techniques used by vision directed perceptive user interfaces are facial expression recognition, head and hand tracking, eye tracking, pose estimation, and gesture recognition.

A. Role of Robotics

Over time, robots have captured significant eminence and become an integral part of our lives in many ways. Not for a reason, they shared our tasks, besides, they bestowed humanity with great ease. Robots have now part of almost every domain: rescue robots [3], bomb diffusing [4], military [5], chemical and explosive material handling [6], and robotic surgeries [7], to name a few.

Robots have now officially incorporated into health-care profession. In [7], a humanoid robotic hand has been developed to assist in urology surgeries. It has a wearable interface, which gives it the flexibility to be operated remotely. Statistics show they are growing each year. Some sources manifest that the worldwide market for medical robots is projected to reach 11.44\$ billion by 2025 [8]. This substantial growth speaks to the impact of robotics in shaping future technological advancements. In the modern day, robotic arms have many use cases, and they are performing magnificently, and so precisely that would have taken humans much longer time and efforts for the same deed.

B. Internet of Robotic Things

Internet of Things (IoT) simply integrates the people, process, device, and technology with sensors and actuators. IoRT is an emerging concept which blends the IoT and robotics, more precisely cloud robotics [9]. Cloud computing is a system that allows a user to access it across continents, with sufficient security conditions, and upon request. Cloud robotics involves cloud computing, cloud storage, and a few other internet technologies revolving around robotics, and the fusion of cloud robotics and IoT gives us an IoRT [10]. The pertinent profit that IoRT involves is the capacity to provide a significant amount of data to the devices associated with the robot without the need to deposit that information directly in the system's memory and the option to control the robot from anywhere around the globe [11].

II. Related Works

In [8] a gesture-controlled robotic arm has been presented, comprising a claw, a flex sensor, a gyroscope, servomotors, and a microcontroller. It is capable of rotating in left, right, up and down directions. Flex sensors contribute in the movement of arm joints, and gyroscope play its part in the motion of arm base. However, this system involves a limitation of not working remotely or wirelessly. A cluster of wires will always be a part of this apparatus causing mess, an increased signal-to-noise ratio (SNR), and the requirement to handle the wires if a system is intended to be operated from a distance of few meters. Also, the flex sensor comes with a bunch of setbacks including limited range of motion, wear and tear, cost and complexity, and calibration issues as well [12]

A design of vision-based hand recognition system has been presented in [13], replacing flex sensor with a swiftly expanding computer vision technology. The framework for this model includes a gesture detecting algorithm and a camera to feed a video to those algorithms for processing. Whereas, a robotic hand enclosing a servo for their corresponding finger, and a microcontroller constitutes a hardware setup. The computer vision segment of this framework has been proceeded in MATLAB, for which a high computational power is required to run and it does not function effectively on C/C++ as Python implementations and their dedicated development environment use to do [14, p. 264]. Besides, this system is also constrained to work wirelessly.

The main difficulty in the implementation of computer vision based hand recognition system stems from the hardware perspective. The available solutions come with limitations that include wear and tear, wearability, and cumbersome issues [8]. Although [13] has implemented the said system which comes with the limitation of being locally operated. In terms of software, it utilizes MATLAB, whereas there are more optimal choices available, such as Python, which is a great preference for building computer vision frameworks and does not require intensive computing power. The approach advocated in this research study is comprised of the following objectives.

- 1) To equip the system with a powerful gesture recognition model using Python that required minimal computing power
- 2) To transmute the system into an IoRT architecture by a procedure to transmit real-time video wirelessly
- 3) To eradicate the need for IoT boards, replace them with a robust video transmission system

rather than sensors or gesture data, which causes latency

- 4) To embrace robotic hand control by deploying a Cortex M0+ processor-based Raspberry Pi Pico board with much higher speed and accuracy
- 5) To forge a system capable of making the robotic hand mimic the gestures of the human hand in real time through camera from anywhere in the world

III. Methodology and Design

The proposed system has been trifurcated as a computer vision segment, a wireless video transfer, and a hardware section as presented in Figure. 1. In the wireless video transfer portion, a camera is accessed over the internet and fed into the computer vision module, where an algorithm is deployed using Python to sense gestures. Based on these gestures, it gives a binary output. Thereafter, the binary output is transmitted to the Raspberry Pi Pico RP2040 microcontroller via *UART* which is responsible to command each servo motor by translating the incoming serial data.

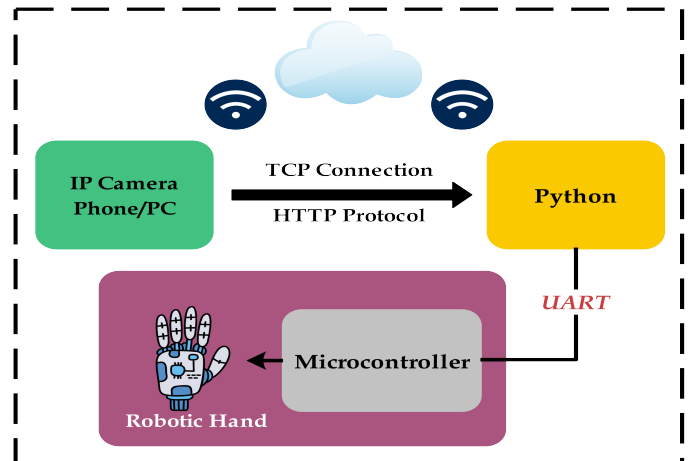


Figure 1: Block diagram of overall proposed system

A. Internet Protocols

There happen to be numerous internet protocols, such as File Transfer Protocol (*FTP*), Simple Mail Transfer Protocol (*SMTP*), Hypertext Transfer Protocol (*HTTP*), Hypertext Transfer Protocol Secure (*HTTPS*), Constrained Application Protocol (*CoAP*) and Message Queuing Telemetry Transport (*MQTT*). Generally, to establish an IoRT framework, *MQTT* protocol is used for its higher speed of transferring data. However, it is not suitable for sending the larger amount of data [15]. In this proposed system, in spite of adopting the *MQTT* protocol, *HTTP* has been utilized for the transmission of live video due to its lower latency characteristic.

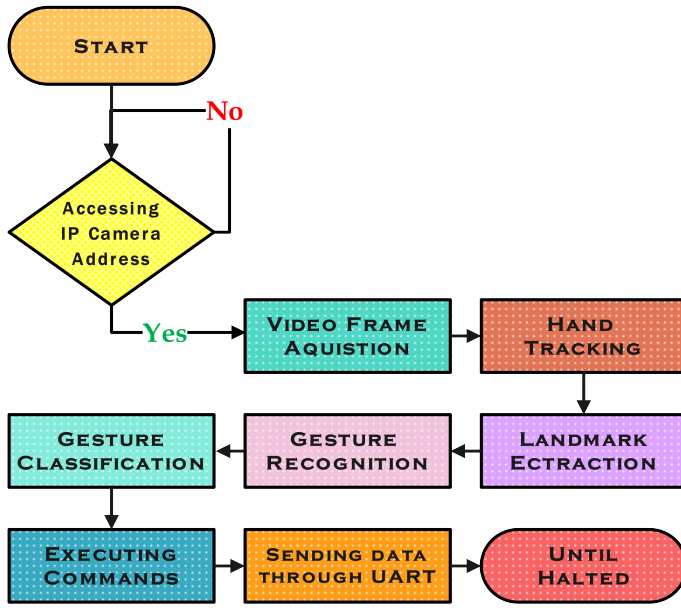


Figure 2: Workflow diagram of computer vision module

B. IoRT Implementation

A flowchart has been presented in Figure. 2, which depicts the sequence of events that took place for the detection of hand gestures in the proposed architecture. Mainly, IoRT architecture can be realized by two approaches. The first approach is named machine-to-cloud (M2C), where two IoT-based microcontrollers are deployed, each functioning as a transmitter or a receiver. The transmitter microcontroller fetches the data from the computer through *UART* and sends it to the IoT cloud. Whereas, the second microcontroller receives that data from the IoT cloud and actuates the robotic hand according to the incoming data. But this kind of framework is most probable to cause latency, communication issues, connectivity challenges, cost concerns, and the complexity of the system [16]. Therefore, to circumvent these hassles, the second approach is introduced in this study, which involves the transmission of video streams over the internet as opposed to deploying hardware that is subjective to latency. For the live video streaming, one can either use an IP camera or turn the phone camera into an IP camera by installing an application called IP Phone Camera [17]. The later strategy has been followed in this research because high-resolution IP cameras are rarely found, or if they are, they are economically unfeasible. This simple approach not only saves cost but also provides ease and convenience. In addition, cell phone cameras are ideal for use in any computer vision application.

C. Computer Vision Module

Computer vision is an area of study which merge the image processing and pattern recognition to bring out the desired results [18]. In the proposed architecture for gesture recognition, a few Python-based libraries have been included. A live video stream is accessed by a computer vision algorithm through an IP camera. The *OpenCV* library has been used to serve this task. Once the incoming video stream is accessed by the *OpenCV* library, it then applies the hand tracking module over it to extract the features and landmarks of the hand. Afterwards, the *MediaPipe* library is utilized for the detection of hands. The *MediaPipe* library provides several pre-trained models, including but not limited to hand detection, face recognition, and pose estimation. Among these, the hand detection model is assigned in this study. The following model comprises palm detection and a hand-landmark unit. Foremost, a palm detection model is implemented on the *OpenCV* treated video stream, which manifests a bounding box. A hand-landmark model is subsequently applied to the cropped image segment, which is characterized by the palm detector unit and reverted by the 21 3D hand key points.

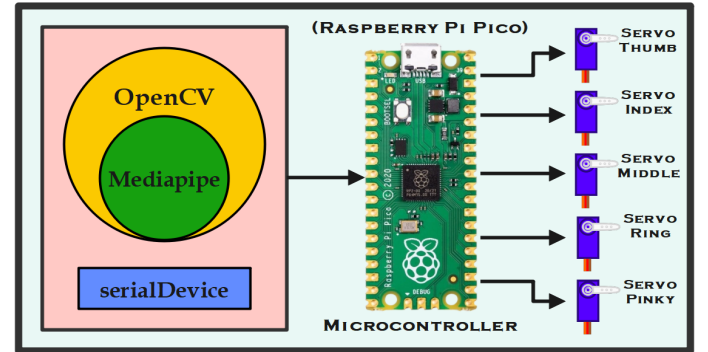


Figure 3: Organization of the proposed system

The keypoints assist in detecting the movement of fingers, which results in the classification of gestures. Subsequently, after gesture detection, it returns a value of 1 for the finger up and 0 for the finger down. These values are then sent to the microcontroller, and based on that, it translates this information and makes the servos rotate accordingly. For this task, it requires a *serialDevice* library which allow the transfer of data from the host computer to Raspberry Pi Pico microcontroller through *UART*. The technical parameters of this serial communication has been gathered in Table. I.

D. Architecture of Robotic Hand

The foundation of any industrial-scale robot is rooted on a robotic arm that comes mostly with a gripper

Table I: Communication Parameters

Communication parameters	
Port No	COM10
Baud Rate	9600
Digits	5

or end effector [8],[12]. However, this does not serve the purpose of this research. Therefore, it necessitates the development of a robotic hand that can emulate human hand gestures. For the conceptual validation, a hardware prototype of a robotic hand has been formulated. It is crafted using polyurethane foam, which has several advantages over polystyrene foam. These include light weight, a high strength-to-weight ratio, flexibility, and reduced noise. The light weight property of polyurethane complements the stable control of fingers. It also offers movement flexibility, which overall improves hand operation and grip. While the servo motors provoke vibrations that cause disturbance, polyurethane foam has the ability to absorb these vibrations as well. For the control of fingers, servo motors with a part number of *SG-90* have been used. It possesses sufficient torque to pull the fingers and has the rotation freedom of 0° to 180° . Therein, the servo motors are placed in spots where they can be fully stretched with ease. A total of five servo motors are being used for the robotic hand, corresponding to each finger. The circuit layout of the proposed system has been presented in the Figure. 4.

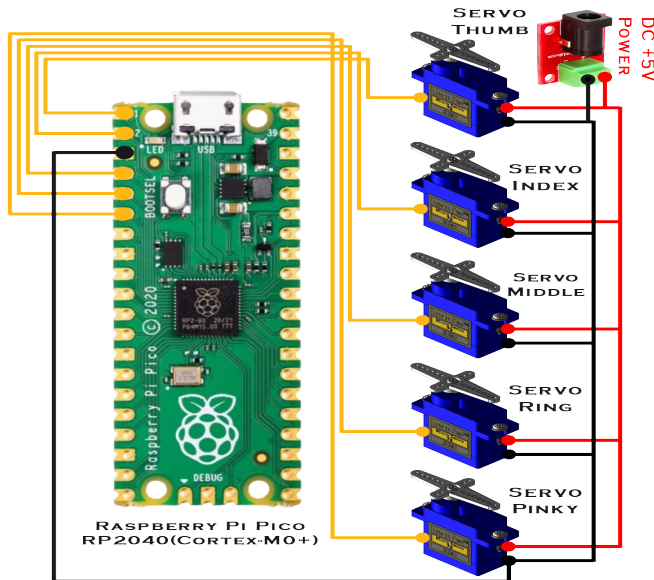


Figure 4: Circuit layout of the hardware

The signal pin of each servo motor is attached to

one of the digital PWM pins of the Raspberry Pi Pico microcontroller board. Specifications of servo motor used in the presented hardware have been presented in Table. II.

IV. Results & Discussion

The proposed system has been first implemented in the Proteus software using Arduino IDE. After getting the desired results, the said system then implemented physically by using Arduino UNO, and Raspberry Pi Pico RP2040 microcontroller board. The testing of hardware has been carried out using two different microcontrollers to not only endorse the simulation work but also have a bench-marking between these microcontrollers. The superiority of Raspberry Pi Pico microcontroller based on technical parameters has been presented in Table. II.

A. Software Results

The simulation of proposed system has been presented In Figure. 5. Arduino UNO is connected to the servos in the same manner as illustrated in Figure. 4 with the Raspberry Pi Pico board. A virtual terminal in Proteus has been used to send and receive data from the microcontroller. It can be noticed that the TX pin of the Arduino board has been connected to the virtual terminal RX, and vice versa. The data coming from the microcontroller is read by the virtual terminal and gets printed on the virtual terminal window as defined in the code. The same is true when the data is being transmitted from the virtual terminal window to the microcontroller board. In the virtual terminal window, data is being transferred in the form of an array. A set of five bits, corresponding to five fingers, is packed inside an array and transmitted from the virtual terminal window. The letter "a" has been added intentionally in the beginning of array to avoid the bits of two separate arrays to get mingled. A loop would be initiated when it detects the letter "a" and starts counting the bits of an upcoming array. When it counts a total of six bits, five values for fingers, plus one of these additional letter "a" the counter gets reset and start counting the next array values. In this way, the microcontroller keeps track of the coming bits and translates them to activate servo motors. When value "1" is detected, it adjusts the angle of the servo and raises the finger, and when value "0" is detected, the finger is lowered. This can also be seen from the servo motor angles in the simulation presented in Figure. 5. Motors are changing the position of their horn arms based on the PWM signal generated by the microcontroller and the values transmitting from the virtual terminal window to the microcontroller.

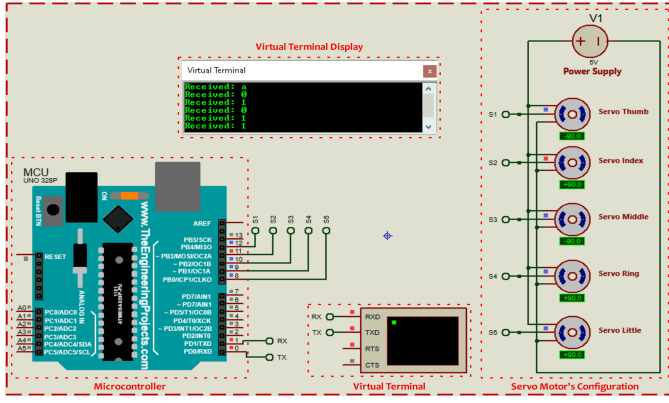


Figure 5: Proteus simulation

B. Hardware Results

Lucrative experimental results of the proposed system has been obtained during the testing of the system hardware. Smooth and accurate movements of the robotic hand have been deduced from the gestures sent through the IP camera of the mobile phone. It has also been noticed that with the Arduino UNO microcontroller, there has been some lag in translating serial commands coming from the PyCharm (Python Interpreter) into the servomotor output presented in Figure. 6(b). The responsiveness and end-to-end latency are assessed by multiple factors, such as CPU, GPU, and internet speed [18], which establish the proposed system's smoothness and accuracy. It is also determined by the response time of Raspberry Pi microcontroller in Figure. 6(a), whose output response was recorded using time stamps in comparison to the Arduino microcontroller, which was faster comparatively.

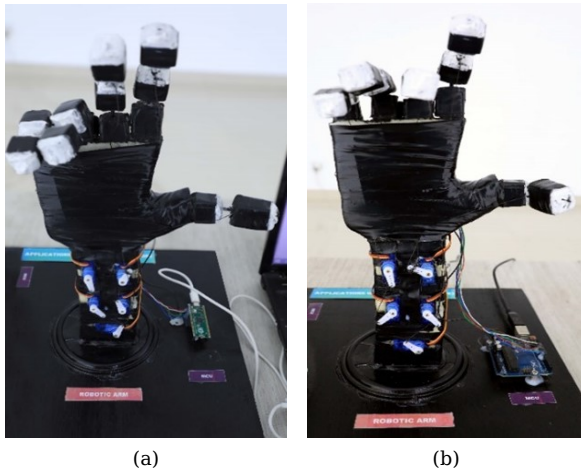


Figure 6: Hardware prototype of the robotic hand

The specifications of both microcontrollers are detailed in Table. II.

Table II: Hardware Specification

Raspberry Pi Pico	
Processor	ARM Cortex M0+
Architecture	32-Bit
Core	2
Clock Frequency	Upto 133MHz
Digital & Analog I/O	26 & 4
Flash Memory	2MB
RAM	264KB
Arduino UNO	
Processor	AVR Microchip
Architecture	8-Bit
Core	Single
Clock Frequency	16MHz
Digital & Analog I/O	14 & 6
Flash Memory	32KB
RAM	2KB
Motor	
Motor type	Servo SG90
Operating Voltage	5V
Torque	1.8 Kg-cm
Motor Weight	9 gram
Operating Angle	0°-180°

The real-time execution of the proposed system has been presented in Figure. 7. An internet connected cell phone camera has been transformed into an IP camera through an application available on the internet named as "IP phone camera". This IP camera records the real time gestures displayed in front of it, and the live stream is accessed by the Python code running in the PyCharm IDE. This live video undergoes through a series of events demonstrated in Figure. 2 to extract the gestures. On the basis of these gestures, it delivers the binary array corresponding to the fingers open or closed. Hereafter, the binary array gets transferred to the microcontroller through *UART* and is handled by the microcontroller. The microcontroller is responsible to translate that binary array according to the instructions fed and changes the servo angle, which ultimately moves the fingers.

In Figure. 7(a), a thumb, index finger, and middle finger of a robotic hand are mimicking the gesture being recorded by the mobile phone camera. In Figure. 7(b), The index, middle, and ring finger of a robotic hand are mimicking the gesture being recorded by a mobile phone camera. All the robotic arm gestures

were in accordance with the gestures fed by the video stream. In Figure 7(c), a detailed close up of cellphone IP camera has been shown.

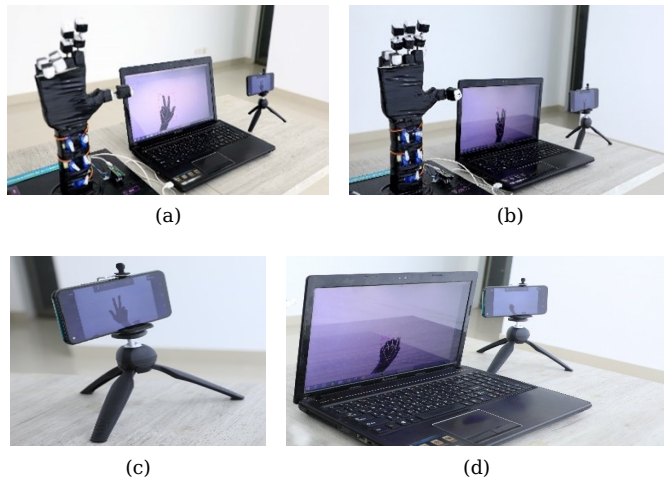


Figure 7: Proposed IoRT architecture operation in real time

Similarly in Figure 7(d), the wireless transmission of the video from the mobile IP camera to the laptop is apparent, which is being accessed through the *OpenCV* library. Despite the mobile camera flipping the incoming video stream, the *OpenCV* library provides the flexibility to mirror the fetched video stream with a few lines of instructions as presented.

V. Conclusion

The suggested approach successfully incorporates a computer vision algorithm with a robotic hand in an IoRT fashion. This provides an intuitive and responsive solution for workers exposed to hazardous conditions while upholding their safety and well-being. The experimental validation involves the utilization of Arduino UNO and Raspberry Pi Pico microcontrollers unitarily for benchmarking their performances and evaluating an IoRT architecture by substituting the M2C framework. In contrast, M2C relies on two IoT-based microcontrollers, which results in increased costs, a more complex framework, and higher latency.

VI. Future Works

Albeit the proposed system can be a precursor to telesurgeries, the futuristic work involves improvements in the robotic arm and gesture detection by streamlining the software algorithms capabilities. Where they could be capable of responding to complex gestures and the robotic hand itself with the extension of arms and equipping it with claws and surgical tools.

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